



Fig. 14: HyperTerminal connection



Fig. 15: COM port settings

To test the source code, first flip switch S7 to 'program' position. Now connect the RS232 serial cable between the PC and the development board. Burn the hex code into the MCU.

Flip switch S7 to the USART position to introduce the MAX232 level converter into the circuit. Open the HyperTerminal window on the PC and set up a new connection as follows:

1. Enter a name for the new connection and click 'Ok' button as shown in Fig. 14.

2. Select the COM port and configure its properties as shown in Fig. 15.

If you have successfully created a new connection, you will see a welcome screen with "Press any key to continue" message. If not, press reset switch S5 once and then check the terminal window.

Follow the instructions on the screen and navigate through the several menu options to control the LEDs,

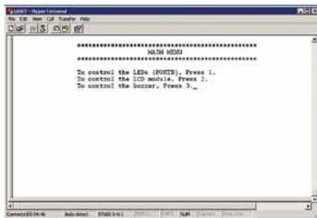


Fig. 16: Main menu window

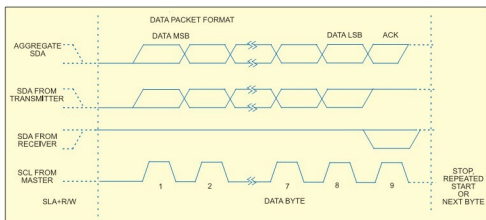


Fig. 17: Data packet format in TWI communication



Fig. 18: TWI module user interface

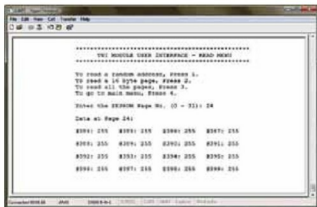


Fig. 19: TWI read menu

LCD module and buzzer on the development board (refer Fig. 16).

Example 12: Two-wire serial interface

The two-wire serial interface (TWI) protocol allows system designers to connect many devices to an MCU using only two bidirectional bus lines: One for

clock (SCL) and the other for data (SDA).

The two.c code written to test the TWI module allows the user to write and read data at random location as well as a page of the externally connected AT24C04 EEPROM device. In AT24C04, the memory is organised as 512x8 memory units. So you have 512 address locations in the EEPROM, each with a 1-byte data word length.

Data packet format.

All the data packets transmitted on the TWI bus are nine-bit long, consisting of one data byte and an acknowledge bit (refer Fig. 17). During data transfer, the master generates the clock and START and STOP conditions, while the receiver is responsible